

DATA 202 · FINAL PRESENTATION

Statistical Analysis of Customer Spending at Starbucks.

*"Every great solution starts with a good cup of
coffee."*

PRESENTED BY

Naman Sudan · Priya Venugopal Nair

Sanjana Vivek Phalke · Varuna Vyomessh



FINAL SUBMISSION · MAY 2026

A behavioral map of 100,000 Starbucks orders.

Why customer spend matters - and what we can ask of this data.

100,000

orders

Order-level transactions, no missing values

14,988

unique customers

Repeat-purchase history captured

500

stores

Urban, suburban & rural locations

20

variables

Channel, cart, customizations, satisfaction...

WHY THIS MATTERS

Spend per order is the single revenue lever that ties together channel strategy, loyalty design, and the in-store experience.

With zero missing values and twenty rich behavioral variables, this dataset lets us go beyond averages to ask what actually *drives* a higher ticket.

What factors drive customer spending at Starbucks?

A central question, broken into five DATA 202 lenses.

Which segments spend more? Channel, rewards, age, store type.

How do cart size and customizations relate to spend? Correlation & covariance.

Are Mobile and Rewards differences statistically real? Two-sample & paired tests.

How likely are high-spender orders - and who places them? Probability & Bayes.

Can linear algebra and optimization model spend? Normal equations & gradient descent.

The shape of an average order.

\$14.87

mean total_spend

\$14.17

median

\$5.51

std. deviation

\$7.34

interquartile range

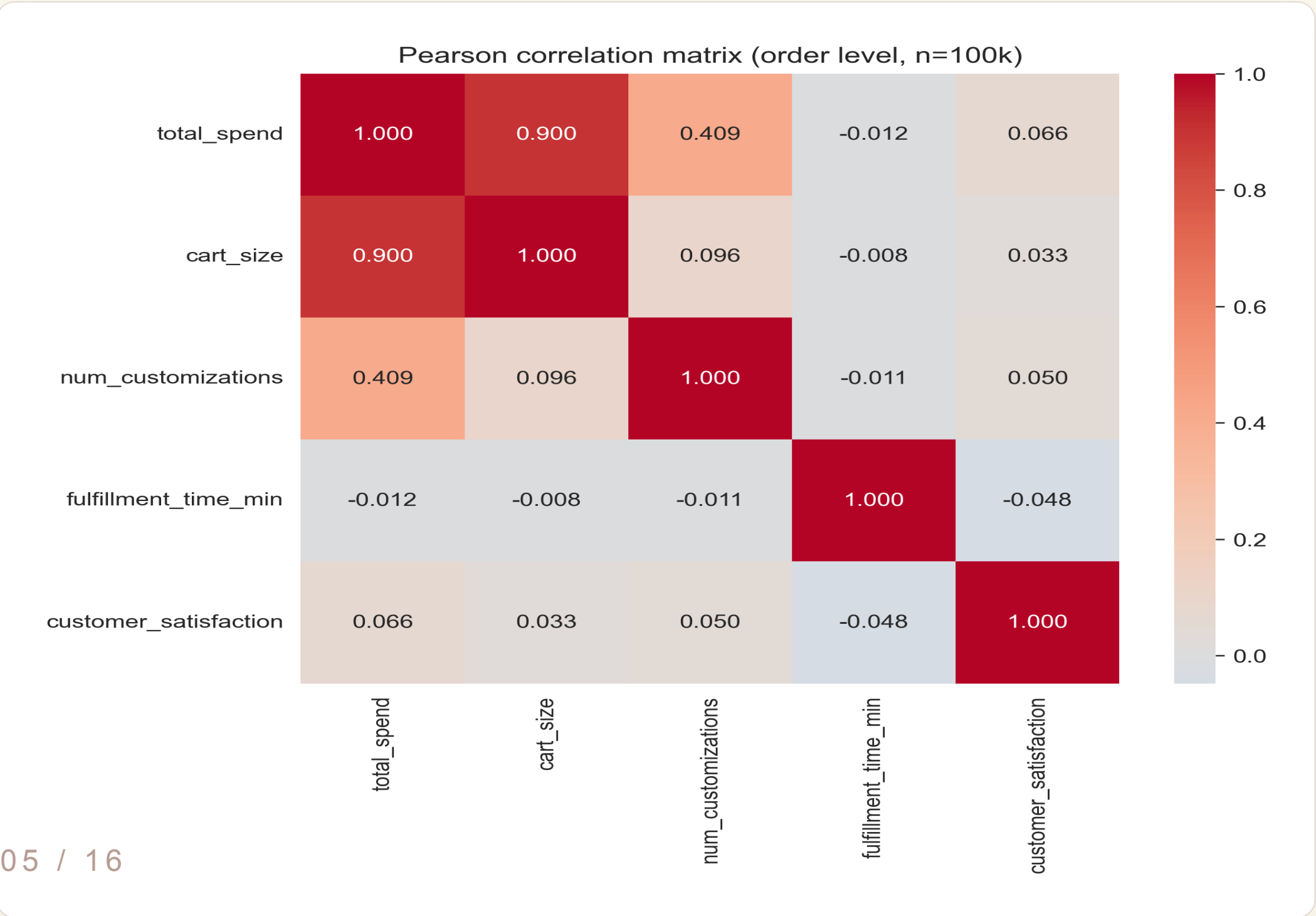
Outliers above \$29.19 · just 1.42% of orders.

MEAN SPEND BY SEGMENT

Segment	Mean spend
Mobile App	\$18.08
Drive-Thru	\$12.48
In-Store Cashier	\$12.51
Kiosk	\$12.49
Rewards member	\$15.72
Non-rewards	\$14.09

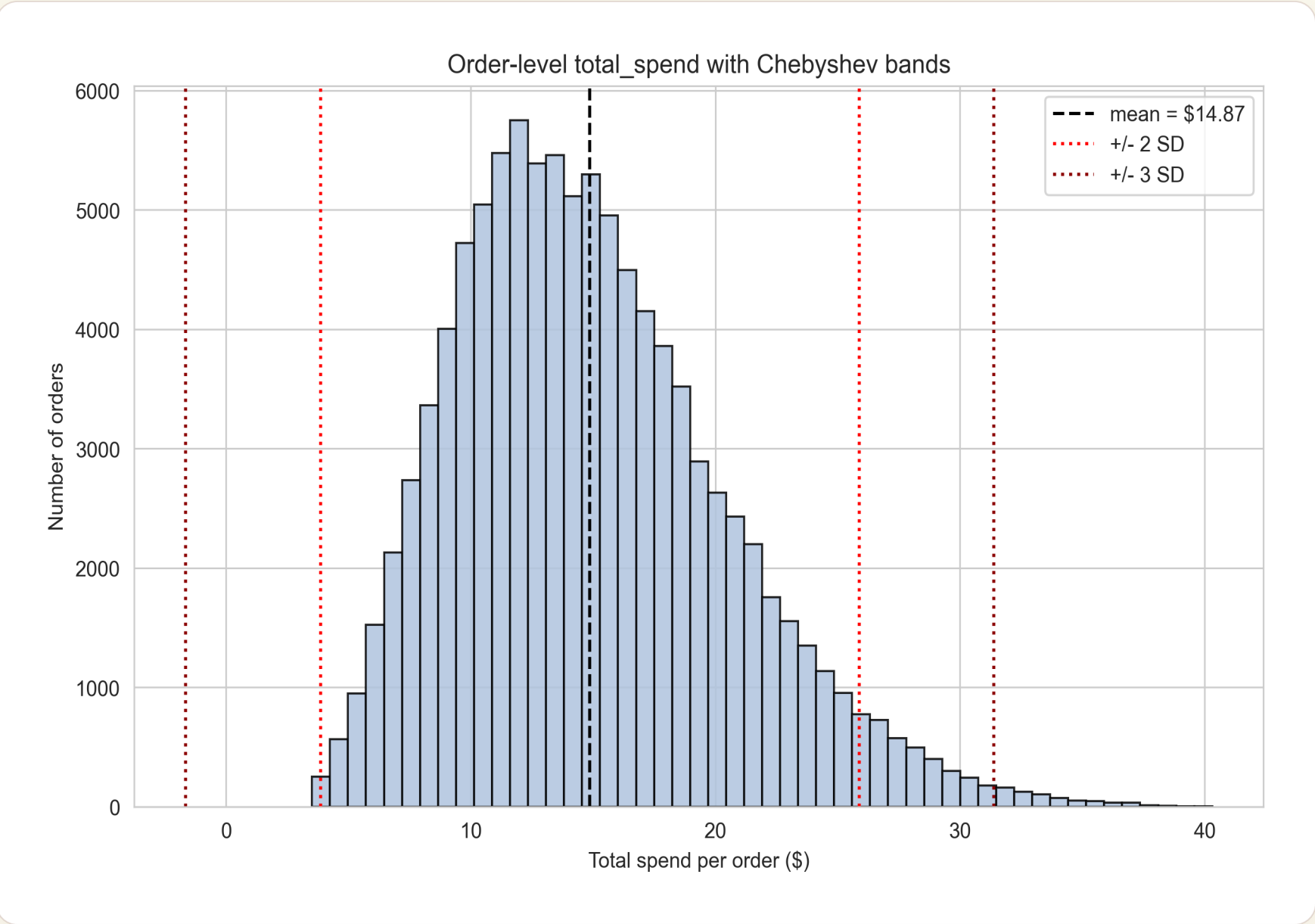
Mobile App leads every other channel by ~\$5.58.

Cart size, not customizations, is the dominant driver.



Cart size is near-deterministic for spend. Customizations matter, but only secondarily. Fulfillment time and satisfaction are essentially uncorrelated with spend.

Tighter than Chebyshev, but not quite normal.



95.83%

of orders within ± 2 SD
Empirical

99.34%

within ± 3 SD
Empirical

$\geq 75\%$

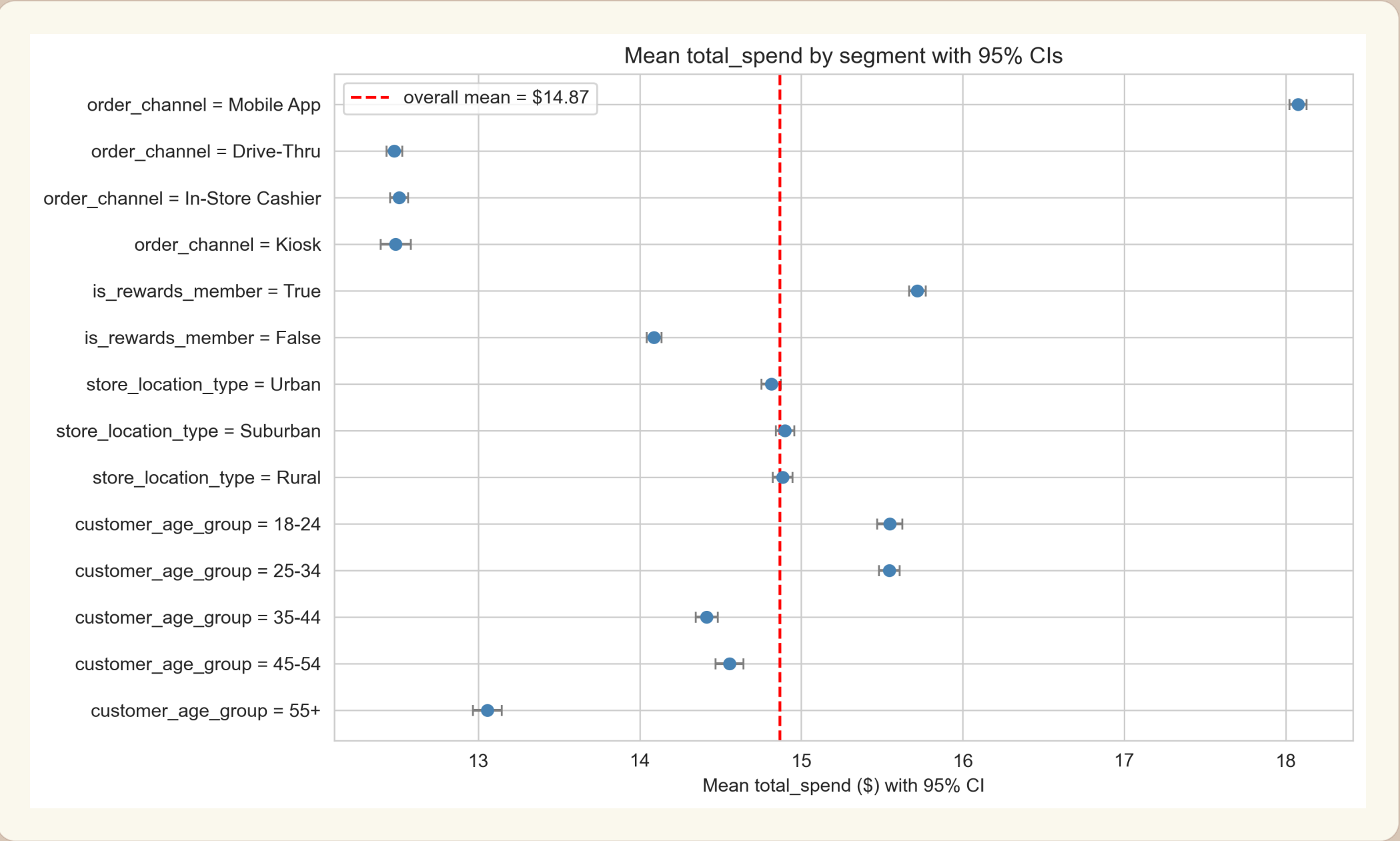
Chebyshev floor at 2 SD
Distribution-free

right-skew

moderate
Normal underestimates the right tail

Because normality is imperfect, we use **empirical probabilities** - not the normal approximation - for tail events.

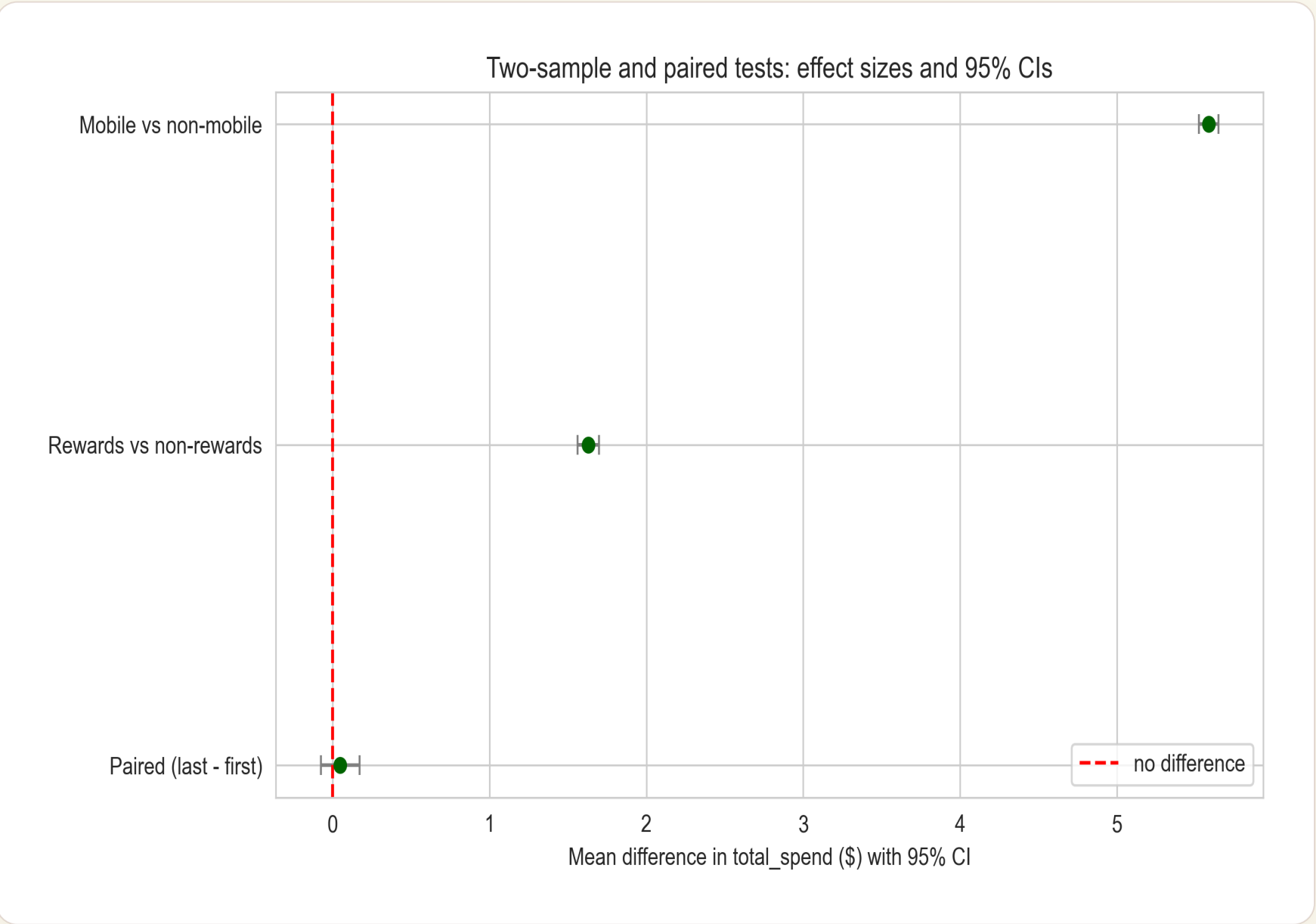
Mobile and Rewards bands don't overlap with the rest.



Segment	95% CI
Overall mean	[14.83, 14.90]
Mobile App	[18.02, 18.13]
Rewards	[15.67, 15.77]
Non-Rewards	[14.04, 14.13]

With $n \approx 100,000$, intervals are narrow. The Mobile band sits well above every other channel - previewing the hypothesis tests.

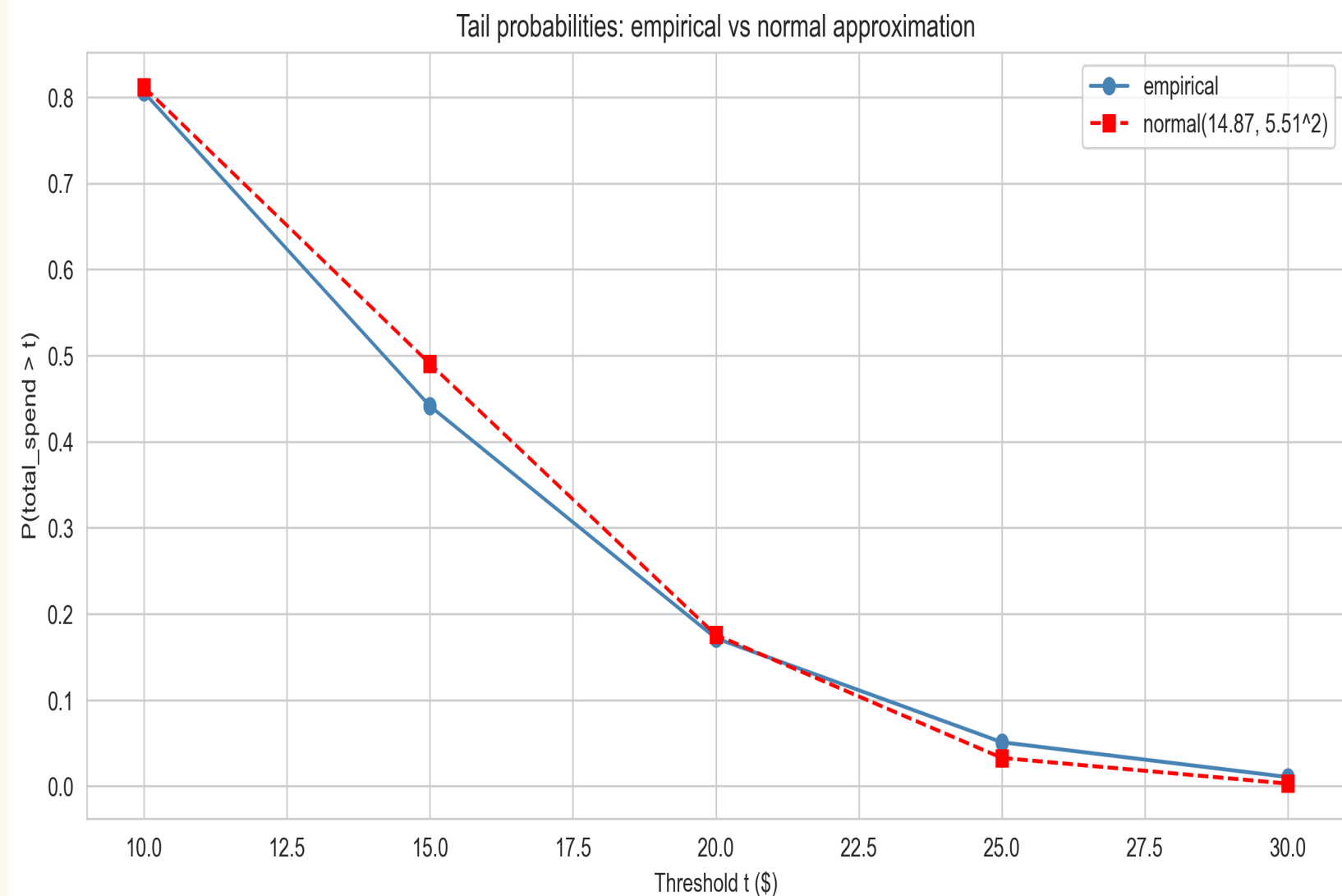
Four tests, four verdicts.



Test	Diff	Stat	p	Decision
Mobile vs Non-Mobile	\$5.58	t=175.91	≈ 0	REJECT H ₀
Rewards vs Non-Rewards	\$1.63	t=47.08	≈ 0	REJECT H ₀
First vs Latest (paired)	~ 0	-	0.44	FAIL TO REJECT
>\$20 high-spender Rewards vs Non	+8.6pp	z = 35.92	≈ 0	REJECT H ₀

Mobile and Rewards differences are **statistically significant** . Repeat-customer spend is **stable over time**.

High spenders are almost always Mobile.



5.11%

$P(\text{spend} > \$25) \text{ empirical}$

3.29%

Normal approximation
(underestimates tail)

11.27%

$P(\text{spend} > \$25 \mid \text{Mobile})$

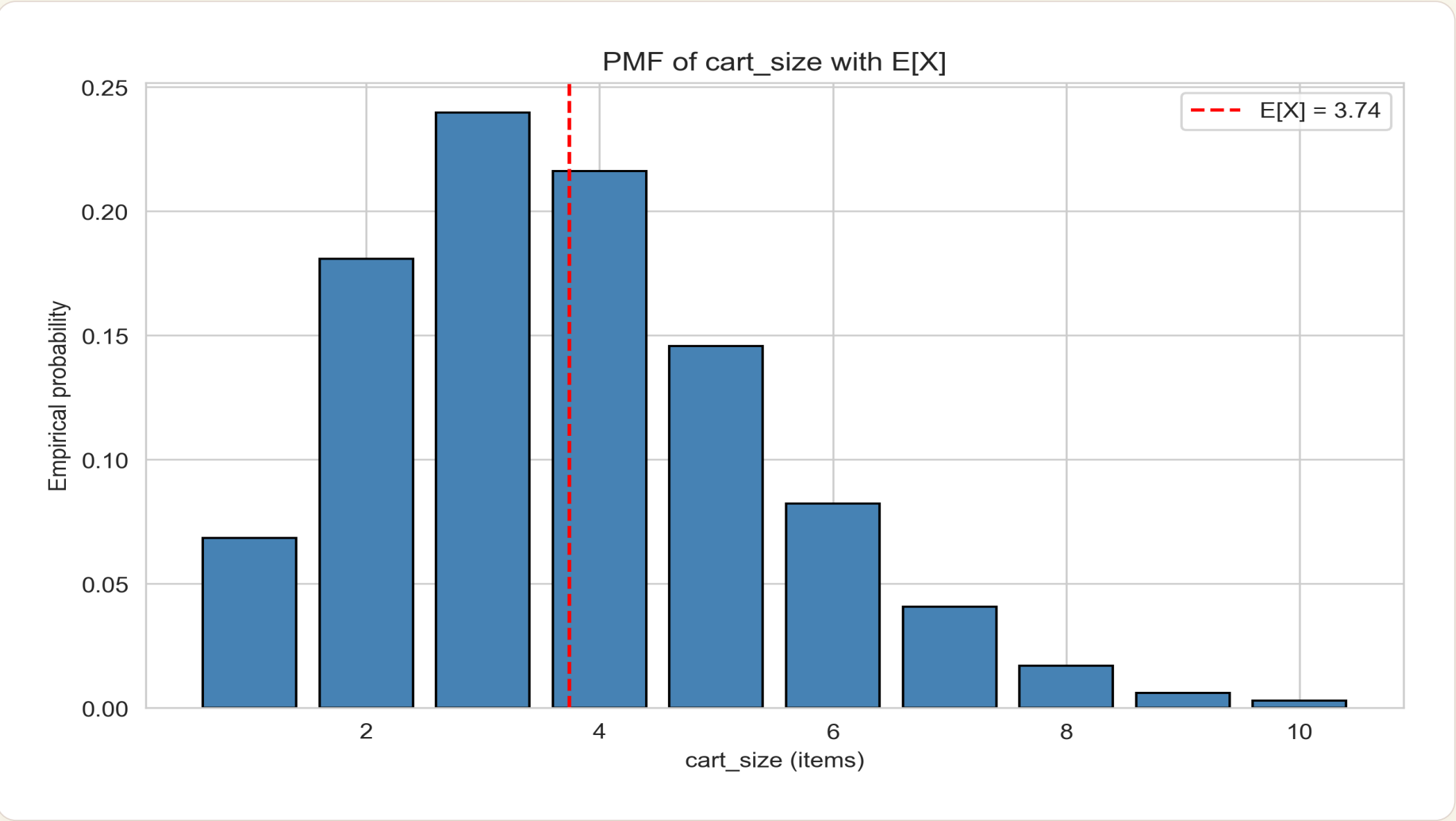
93.85%

$P(\text{Mobile} \mid \text{spend} > \$25) \text{ Bayes}$

And $P(\text{Rewards} \mid \text{spend} > \$25) = 62.74\%$ - high spenders are disproportionately Mobile, Rewards customers.

Cart size, formally.

A discrete RV over $\{1, 2, \dots, 10\}$ - the unit of work behind every dollar of spend.



3.7415

$E[\text{cart_size}]$ · expected items per order

2.8826

$\text{Var}(\text{cart_size})$

Mode at 3 items, right-skewed tail to 10. Treating cart size as an RV ties the descriptive picture directly to the linear model on the next slide.

A 3-feature model that explains 95% of spend.

$\hat{y} = \beta_0 + \beta_1 \cdot cart_size + \beta_2 \cdot customizations + \beta_3 \cdot mobile_app$

intercept (β_0)

+ 2.16

cart_size

+ 2.69 / item

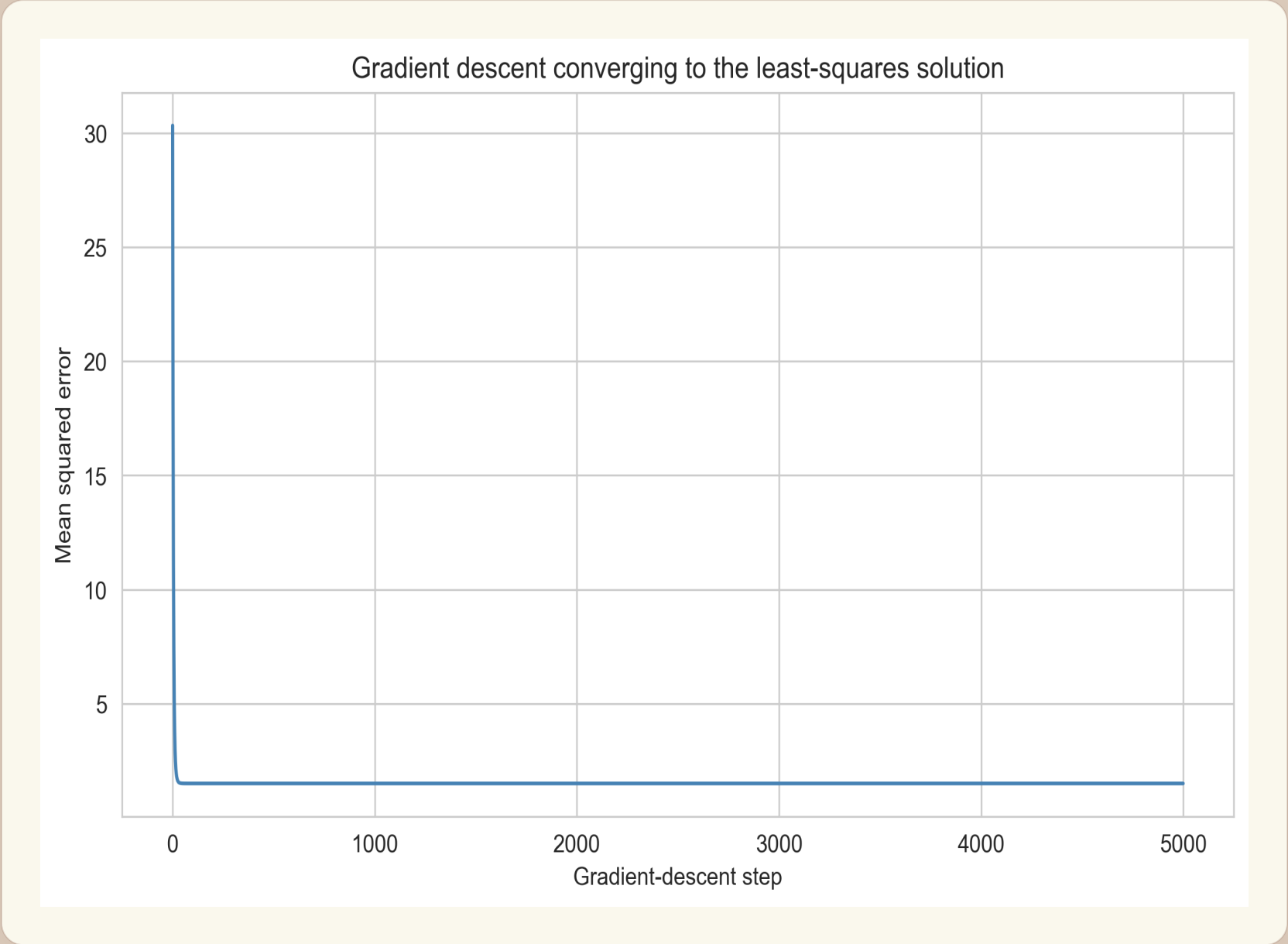
num_customizations

+ 0.92 / mod

mobile_app

+ 2.33

$R^2 = 0.9505$
95% of variance in total_spend explained



Solved two ways - closed-form via the normal equations, iteratively via gradient descent. Both converge to the same coefficients.

From coefficients to revenue levers.

Mobile App is the largest lever.

+\$5.58 vs every other channel. 93.85% of high-spender orders come through Mobile.

Cart size is the strongest predictor.

$r = 0.900$ with spend. Each extra item adds \$2.69 in the linear model.

Rewards is smaller but real.

+\$1.63 per order, statistically significant. Rewards members are 62.74% of high spenders.

High spenders are a Mobile phenomenon.

Channel mix - not just menu - is what unlocks above-\$25 orders.

From notebook to production.

01

Batch refresh

Recompute descriptive stats and refit the model on a regular cadence.

02

Online scoring

Predict expected spend at order time to drive personalized upsells.

03

Drift monitoring

Track Mobile share, Rewards effect, and high-spender probability over time.

04

Scale

Re-fit by store or region; same recipe, finer granularity.

The same three coefficients that explained 95% of variance in our notebook can power live recommendations in the Starbucks app.

Honest about what we can't claim.

Sample representativeness. Kaggle data may not cover every real Starbucks store or geography.

Independence. Repeat customers weaken strict independence assumptions in t-tests.

Normality testing. Shapiro–Wilk uses a 5,000-row subsample because of its sample-size cap.

Tail estimation. The Normal approximation underestimates the right tail of spend.

Paired test. Compares first vs latest order, ignoring unequal time gaps between them.

Model simplicity. Three predictors only - intentionally course-aligned, not state of the art.

Mobile App and cart size drive spend.

Four findings that hold across every method we ran.

Mobile App and cart size are the dominant drivers of spend per order.

Empirical tail probabilities are more honest than the normal approximation for high spenders.

Rewards membership is a smaller but statistically significant lift.

Methods align with DATA 202 - descriptives, correlation, hypothesis tests, Bayes, RVs, linear algebra, gradient descent.

Same conclusion from five different lenses - that's what makes the result trustworthy.

- END OF PRESENTATION -



Thanks *a latte.*

QUESTIONS · DISCUSSION